ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** a simplified representation of a system that keeps what matters for the question	Models leave out unimportant details.
2	**(c)** a single kind of particle with fixed composition	Pure substances have fixed composition.
3	**(d)** Antoine Lavoisier	Lavoisier.
4	**(a)** electron	Cathode ray experiments.
5	**(a)** an approximate answer is enough for a decision	Orders of magnitude.
6	**(d)** a heterogeneous mixture whose particles settle on standing	Large particles.
7	**(c)** 4	Four atoms.
8	**(d)** protons	Z = p.
9	**(a)** a law describes what happens while a theory explains why	Different roles.
10	**(b)** pencil does not dissolve in the solvent	Ink would move.
11	**(a)** negatively charged ion	Gains electrons.
12	**(a)** 1	Gains 1 to make 8.
13	**(b)** a model	A visual model of matter.
14	**(a)** distilled water	Single compound.
15	**(a)** Joseph Proust	Proust.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Simplified representation of a real system (1); Ball-and-stick model of a molecule (1).	1+1
17	Homogeneous: uniform, e.g. salt solution (1); Heterogeneous: non-uniform, e.g. sand and water (1).	1+1
18	Mass is neither created nor destroyed in a chemical reaction (1); Antoine Lavoisier (1).	1+1
19	It could not explain the scattering of alpha particles (1); It had no nucleus (1).	1+1
20	It is a deliberate assumption to simplify (1); The model is accurate enough for the purpose (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Choose features important to the question (1); Ignore minor details such as air resistance for a falling ball (1); Assumptions make the problem manageable (1).	1+1+1
22	Heat the mixture: ammonium chloride sublimes (1); Add water to the residue, filter sand (1); Evaporate the filtrate to get salt (1).	1+1+1
23	Mass of reactants = 22.0 g (1); Mass of CO2 formed = 22.0 g (1); Mass before equals mass after (1).	1+1+1
24	Alpha particles directed at thin gold foil (1); Most passed; some deflected; very few rebounded (1); Nucleus small, dense and positive; mostly empty atom (1).	1+1+1
	OR: Units were confused (litres and gallons or pounds and kilograms) (1); Wrong quantity of fuel was loaded (1); Care with units is vital (1).	1+1+1
25	12/100 x 250 (1); = 30 g (1); Water = 220 g (1).	1+1+1
	OR: Hard, brittle crystalline solids (1); High melting and boiling points (1); Conduct electricity when molten or dissolved (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Question from observation (1); Hypothesis and prediction (1); Fair test experiment (1); Data collection and analysis (1); Conclusion and further questions (1).	1+1+1+1+1
	OR: Pure: fixed composition (1); Mixture: variable (1); Gold: pure; air, sugar solution, brass: homogeneous (1); Milk: colloid, treated as heterogeneous (1); Properties differ (1).	1+1+1+1+1
27	Take both solutions in a flask with a small tube (1); Weigh the set-up (1); Mix: white precipitate of $BaSO_4$ forms (1); Weigh again: mass is the same (1); Conclusion: mass is conserved (1).	1+1+1+1+1
	OR: Thomson: plum pudding (1); Drawback: could not explain alpha scattering (1); Rutherford: nucleus with revolving electrons (1); Drawback: instability (1); Bohr: fixed shells (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** negative	Negative charge.
2	**(b)** 10000 litres	0.5 x 20000.
3	**(c)** fog	Water droplets in air.
4	**(c)** 8	Eight atoms.
5	**(a)** protons plus neutrons	A = p + n.
6	**(c)** a general principle that applies in every process	A fundamental law.
7	**(a)** moves farther up the paper	Greater solubility.
8	**(a)** NaCl	1:1.
9	**(d)** chemically unreactive	Stable.
10	**(d)** describes a regular pattern observed in nature	Laws describe regularities.
11	**(a)** brass	Alloy has uniform composition.
12	**(b)** equals the total mass of the products	Mass is neither created nor destroyed.
13	**(b)** Rutherford	Gold foil experiment.
14	**(a)** refine the model	Science corrects itself.
15	**(d)** smaller than the pores of the paper	Colloidal particles are small.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Separation by spinning at a high speed (1); Cream from milk, blood cells from plasma (1).	1+1
17	Number of atoms in a molecule (1); 2, 3 and 4 (1).	1+1
18	Protons = electrons = 11 (1); Neutrons = 12 (1).	1+1
19	Examine assumptions and measurements (1); Revise the model and test again (1).	1+1
20	Mass of solution = 100 g (1); 25 per cent (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Isotopes: same Z, different A (1); Isobars: same A, different Z (1); Isotopes have the same chemical properties (1).	1+1+1
22	Mass = 2.5 kg (1); = 2500 g (1); Density used (1).	1+1+1
23	Saturated: no more solute dissolves (1); Unsaturated: can dissolve more (1); Heat or add solvent to make unsaturated (1).	1+1+1
24	Atom: smallest particle of an element (1); Molecule: group of atoms bonded (1); Atoms of noble gases exist alone; others usually as molecules (1).	1+1+1
	OR: Protium: 1 proton, 0 neutron (1); Deuterium: 1 proton, 1 neutron (1); Tritium: 1 proton, 2 neutrons (1).	1+1+1
25	One breath about 0.5 litre (1); About 20000 breaths a day (1); 0.5 x 20000 = 10000 litres (1).	1+1+1
	OR: Dissolve in hot water to a saturated solution and filter (1); Cool slowly (1); Pure crystals form and are dried (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Instrument limits (1); Human errors like parallax (1); Repeat measurements (1); Take the average (1); Use better instruments (1).	1+1+1+1+1
	OR: Spot above the baseline (1); Solvent rises by capillary action (1); Components move at different rates (1); Compare heights (1); Used for dyes and pigments (1).	1+1+1+1+1

27	Oxygen = 8/3 x 18 = 48 g (1); Compound mass = 66 g (1); Compound is carbon dioxide (1); Ratio fixed by law of constant proportions (1); Conservation of mass verified (1).	1+1+1+1+1
	OR: K: p 19, n 20, e 19 (1); S: p 16, n 16, e 16 (1); F: p 9, n 10, e 9 (1); Atomic number identifies the element (1); Mass number gives nucleons (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 18 u	2 + 16.
2	**(b)** 23	11 + 12.
3	**(c)** burning of magnesium	New substance formed.
4	**(c)** homogeneous mixtures of metals	Metal solutions.
5	**(d)** $CaCl_2$	Ca2+ and Cl-.
6	**(c)** atomic number but different mass numbers	Same Z.
7	**(b)** explains many observations and makes testable predictions	Theories explain and predict.
8	**(a)** heterogeneous	Components are visible.
9	**(d)** 44 g	12 + 32 = 44.
10	**(a)** James Chadwick	1932.
11	**(b)** units must be used carefully and consistently	Importance of units.
12	**(a)** centrifugation	Spinning.
13	**(b)** 44 u	12 + 32.
14	**(a)** 18	35 - 17.
15	**(a)** new evidence required changes	Models are revised.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	By sharing of electron pairs between atoms (1); H_2 or Cl_2 (1).	1+1
17	Iodine-131 for goitre (1); Cobalt-60 for cancer treatment (1).	1+1
18	Law describes a regular pattern (1); Theory explains why it happens (1).	1+1
19	Maximum mass of solute that dissolves in 100 g of solvent at a given temperature (1); Generally increases with temperature (1).	1+1
20	A pure compound always contains elements in a fixed ratio by mass (1); Water: H : O = 1 : 8 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Fraction of oxygen = 8/9 (1); Mass = 8/9 x 27 (1); = 24 g (1).	1+1+1
22	He has 2 protons but mass 4 u (1); Neutral particles of mass 1 u account for the rest (1); Helium: 2 protons + 2 neutrons (1).	1+1+1
23	Treated matter in stars as ionised gas (1); Applied known laws of physics and chemistry (1); Explained the different spectra of stars (1).	1+1+1
24	Alcohol = 15/100 x 200 = 30 mL (1); Add to some water in a measuring cylinder (1); Make up to 200 mL (1).	1+1+1
	OR: Low melting and boiling points (1); Poor conductors (1); Often soluble in organic solvents (1).	1+1+1
25	Orbiting electrons should lose energy and spiral into the nucleus (1); It could not explain the stability of atoms (1); Bohr introduced fixed energy shells (1).	1+1+1
	OR: Accuracy: closeness to the true value (1); Precision: closeness of repeated readings (1); Example: 15.2 and 15.3 cm are precise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Ball and stick model (1); Atomic models (Dalton to Bohr) (1); Ideal gas or frictionless motion (1); New evidence leads to revision (1); Models are useful, not final (1).	1+1+1+1+1

	OR: Dissolve salt, mix chalk and milk in water (1); Observe clarity (1); Filter each (1); Shine light (1); Observe settling (1).	1+1+1+1+1
27	Matter made of atoms (1); Atoms of an element identical (1); Atoms of different elements differ (1); Atoms combine in simple ratios (1); Atoms are neither created nor destroyed; later: atoms divisible, isotopes exist (1).	1+1+1+1+1
	OR: Thin gold foil, alpha source, fluorescent screen (1); Most passed straight: empty space (1); Some deflected: positive charge (1); Few rebounded: small dense nucleus (1); Nuclear model (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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